Stephanie J. Spielman, PhD

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Education

The University of Texas at Austin *Austin, Texas, USA*
Ph.D., Integrative Biology *2016*

Brown University *Providence, RI, USA*
B.S., Biology with Honors *2010*

Academic/Scientific Positions

Alex's Lemonade Stand Foundation *Bala Cynwyd, PA, USA*
Data Scientist, Childhood Cancer Data Lab *2022-2026*
- Develop open-source tools, bioinformatics analysis pipelines, and training workshops to accelerate pediatric cancer research

Rowan University *Glassboro, New Jersey, USA*
Assistant Professor *2018-2022*
- Ran undergraduate research lab focused on methods in molecular evolution and developed undegradute-level courses in Evolutionary Biology and Data Science

Temple University *Philadelphia, Pennsylvania, PA*
Research Assistant Professor *2016–2018*
- Developed research in methods for quantifying protein evolution and taught graduate-level courses in Biostatistics and Evolutionary Medicine

The University of Texas at Austin *Austin, Texas, USA*
Graduate Research Assistant *2011–2016*
- Computational molecular evolution research, including phylogenetic modeling and virus evolution.

The University of Texas at Austin *Austin, Texas, USA*
Graduate Teaching Assistant *2011–2016*
- Supported undergraduate level courses in Biostatistics, Computational Biology and Bioinformatics, and Evolutionary Biology

Honors and Awards

President's Award for Excellence in Innotative Instructional Delivery *Glassboro, NJ, USA*
Rowan University *2021*
- Awarded by the Faculty Center for Teaching Excellence

Outstanding Dissertation Award *Austin, Texas, USA*
The University of Texas at Austin *2016*
- Awarded by the Office of Graduate Studies to the top dissertation among all of science, math, and engineering

Ruth L. Kirschstein National Research Service Award (F31) *Austin, Texas, USA*
The University of Texas at Austin *2015*
- Pre-doctoral fellowship awarded by NIH/NIGMS

Outstanding Teaching Award *Austin, Texas, USA*
The University of Texas at Austin *2014*
- Awarded by the Biology Instructional Office

Selected Publications

1. Hawkins, A. G., Shapiro, J. A., Spielman, S. J., Mejia, D. S., Prasad, D. V., Ichihara, N., Yakovets, A., Gottlieb, A. M., Wheeler, K. G., Bethell, C. J., Foltz, S. M., O'Malley, J., Greene, C. S., & Taroni, J. N. (2026). The single-cell pediatric cancer atlas: Data portal and open-source tools for single-cell transcriptomics of pediatric tumors. *Cell Genomics*, *6*(7). https://doi.org/10.1016/j.xgen.2026.101283
2. Shapiro, J. A., Gaonkar, K. S., Spielman, S. J., Savonen, C. L., Bethell, C. J., Jin, R., Rathi, K. S., Zhu, Y., Egolf, L. E., Farrow, B. K., Miller, D. P., Yang, Y., Koganti, T., Noureen, N., Koptyra, M. P., Duong, N., Santi, M., Kim, J., Robins, S., … Taroni, J. N. (2023). OpenPBTA: The open pediatric brain tumor atlas. *Cell Genomics*, *3*(7), 100340. https://doi.org/10.1016/j.xgen.2023.100340
3. Spielman, S. J., & Miraglia, M. L. (2021). Relative model selection of evolutionary substitution models can be sensitive to multiple sequence alignment uncertainty. *BMC Ecology and Evolution*, *21*, 214. https://doi.org/10.1186/s12862-021-01931-5
4. Spielman, S. J. (2020). Relative model fit does not predict topological accuracy in single-gene protein phylogenetics. *Molecular Biology and Evolution*, *37*(7), 2110–2123. https://doi.org/10.1093/molbev/msaa075
5. Spielman, S. J., & Moore, E. K. (2020). dragon: A new tool for exploring redox evolution preserved in the mineral record. *Frontiers in Earth Science*. https://doi.org/10.3389/feart.2020.585087
6. Spielman, S. J., & Kosakovsky Pond, S. L. (2018). Relative evolutionary rates in proteins are largely insensitive to the substitution model. *Molecular Biology and Evolution*, *35*(9), 2307–2317. https://doi.org/10.1093/molbev/msy127
7. Echave, J., Spielman, S. J., & Wilke, C. O. (2016). Causes of evolutionary rate variation among protein sites. *Nature Reviews Genetics*, *17*(2), 109–121. https://doi.org/10.1038/nrg.2015.18
8. Meyer, A. G., Spielman, S. J., Bedford, T., & Wilke, C. O. (2015). Time dependence of evolutionary metrics during the 2009 pandemic influenza virus outbreak. *Virus Evolution*, *1*(1), vev006. https://doi.org/10.1093/ve/vev006
9. Spielman, S. J., & Wilke, C. O. (2015). pyvolve: A flexible python module for simulating sequences along phylogenies. *PLOS ONE*, *10*(9), e0139047. https://doi.org/10.1371/journal.pone.0139047
10. Spielman, S. J., & Wilke, C. O. (2015). The relationship between dN/dS and scaled selection coefficients. *Molecular Biology and Evolution*, *32*(4), 1097–1108. https://doi.org/10.1093/molbev/msv003